

	Terrestrial data centre	HAPS-based aerial data centre (this work)	Orbital/space data centre
 Location / platform	Ground	Stratosphere (~20 km)	Low Earth orbit (300–2,000 km) Example: ~500 km
 Power source	Grid power + backup	Onboard solar PV + storage	Solar arrays + batteries; eclipse periods
 Heat rejection environment	Atmosphere via HVAC (air/liquid cooling)	Fan-assisted internal circulation with heat rejection to the stratospheric environment	Radiators radiating to space (vacuum)
 Communication latency / coverage	~0 ms (local)	~0.067 ms one-way to ground (~0.13 ms RTT)	~1.67 ms one-way to ground (~3.33 ms RTT at 500 km)
 Maintainability / accessibility	High	Recoverable but operationally challenging	Low; servicing generally difficult or impossible
 Payload / resource constraints	Site, grid and cooling infrastructure	Payload power, mass and thermal envelope	Launch mass/volume, power, radiator area, radiation environment

